

Lohith Prasanna Teja Kakumanu

716-907-7542 | lohithkakumanu483@gmail.com | [LinkedIn](#) | [Portfolio](#) | [Medium](#)

SUMMARY & SKILLS

Senior Data Science and AI Engineer specializing in production GenAI, RAG, NLP, AI safety, anomaly detection, experimentation, cloud ML platforms, AI Agents, Agentic Loop, Harness, ReAct, Context Engineering, Tool Call, A2A, Guardrails, HITL

Core: Python, SQL, R, PyTorch, TensorFlow, scikit-learn, LangChain, FAISS, AWS Bedrock, SageMaker, Lambda, S3, Glue, Redshift, DynamoDB, Azure Blob Storage, Azure Container Apps, Docker, Kubernetes, Git, Power BI, Tableau, DSPy, Postgres, pgvector.

PROFESSIONAL EXPERIENCE

Sr. Data Science Engineer | Orion Innovation × PwC, NJ | Dec 2025 - Present

- **Multi-Turn Intent Resolution:** Designed **stateful** follow-up inference using **semantic continuity, contextual reference resolution, query rewriting, and stale-context control**, successfully resolving **nearly 90%** of designed multi-turn conversational scenarios.
- **Semantic Intent Classification:** Ran controlled experiments using **hard negatives, representation coverage, threshold calibration, and spurious-correlation controls**, improving semantic separability by **34%** and representative intent similarity from **0.58 to 0.71**.
- **Authoritative-Source Anchoring for RAG:** Invented **chunk-level retrieval control** that guarantees designated baseline evidence enters the generation context alongside top-k retrieval, achieving **100% authoritative-source inclusion** across evaluated scenarios.
- **End-to-End AI Application Pipelines:** Built process APIs, ingestion, digest, preprocessing, retrieval orchestration, schema validation, and embedding initialization, eliminating valid-request HTTP 500 failure paths and achieving a 100% pass **rate**.
- **AI Guardrail Framework:** Architected a **domain-neutral AI guardrail framework** that outperformed NVIDIA NeMo Guardrails on 5,500 public cases by 34 **percentage points** in balanced accuracy and 62 **percentage points** in unsafe-request detection, reducing residual risk from 16.7% to 3.5% ($p < 1e-50$).
- **Production RAG Latency Optimization:** Owned end-to-end optimization of a production RAG chat path, reducing the worst-case intent-gate latency ceiling from **32s to 8s (4× faster)** while preserving **fail-closed safety and stage-level observability**.

Data Science Engineer | Fannie Mae, VA | Feb 2025 - Nov 2025

- **Refined Policy Chatbot:** Integrated **LangChain, FAISS, and AWS Bedrock** to reduce over-context retrieval, improving policy Q&A accuracy and model reliability by **45%**; evaluated responses using **factuality, coherence, and retrieval relevance**.
- **Loan Anomaly Detection:** Built **PyTorch deep-autoencoder inference pipelines** on Lambda, EC2, and S3, cutting diagnostic latency by **50%**; statistical exception checks improved audit accuracy by **35%** and reduced manual review by **60%**.
- **Capital Rule Validation:** Applied z-score, threshold, and percentile analysis to detect discrepancies in loan-level capital calculations, improving audit accuracy by 35% and reducing manual review by 60%.
- **Capital Calculation Platform:** Reengineered a production platform across **S3, Redshift Spectrum, Glue, DynamoDB, 15+ internal APIs, and SageMaker**, increasing throughput by **40%**; authored SOPs, runbooks, knowledge articles, gap analyses, and FAQs.

Data Scientist | Crypto Clear, MA | May 2024 - May 2025

- **Enhanced Fraud Detection:** Tuned **Graph Neural Networks (GNNs)** across 1M+ blockchain transactions, reducing fraud-detection anomalies by 25%, improving prediction accuracy by 15%, and increasing suspicious-activity detection by 30%.

Research Assistant | University at Buffalo, NY | Sep 2023 - Oct 2025

- **Robustness Evaluation:** Fine-tuned ViLT on VQA v2, reducing prediction loss by 30% and improving accuracy by 15%; [published](#) research on a novel sampling strategy **for deep-learning robustness evaluation**, improving estimation accuracy by 15% and interpretability across datasets.

Data Scientist | Infosys India Pvt. Ltd., India | Jul 2021 - Aug 2023

- **Auto-Tagging System Implementation:** Implemented **LLM auto-tagging systems**, reducing manual labeling by **60%** and accelerating model training.
- **NLP and GenAI Chatbot Development:** Developed a generative AI chatbot using **advanced NLP techniques**, improving customer interaction and satisfaction by **15%** while reducing support load.
- **Employee Retention Analysis:** Applied **Propensity Score Matching (PSM) and Difference-in-Differences (DiD)** to evaluate retention strategies, improving decision-making accuracy by **15%** and reducing attrition risk.

Data Science & BI Engineer | Infosys India Pvt. Ltd., India | Sep 2019 - Jul 2021

- **Power BI Enhancement:** Led ETL, data modeling, SQL, and stored-procedure optimization, improving data quality by 25% and accessibility by 35%; optimized 10+ Power BI reports using **performance tuning, DAX**, and row-level security, reducing load times by 20% and improving user experience by 15%.

PROJECTS

Applicant Tracking System: Python, GenAI, and LangChain - Improved candidate skill validation and initial-screening success by **45%** while reducing recruiter workload by **30%**.

Crime Hotspot Forecasting: Python, ARIMA, SARIMA, and Bollinger Bands - Engineered time-series models to forecast crime rates, identifying a **15% decrease** in states experiencing 5% or more economic growth.

EDUCATION

M.S. in Industrial and Systems Engineering - University at Buffalo-SUNY, NY; **GPA: 4.0/4.0**; Dec 2024

B.Tech. in Electronics and Communication Engineering - VVIT, India; **GPA: 3.04/4.0**; Sep 2020

RECOGNITION

- **INFY-RISE Award:** Recognized for exemplary leadership and execution on a business-critical project.
- **Infosys Certifications:** Data Science Professional, SSIS Developer, and SQL Professional.
- **Design Council Head, IETE Student Chapter:** Organized **15+ events and workshops**, engaging **200+ students** at VVIT.